

Quiz 3 - Regression - Instructor key

Private release QUIZ03, revision 1. Five one-page forms; 10 minutes; 3 points. Question order: easy / medium / hard.

Use any one form per sitting. The five forms assess the same three items with varied numerical inputs and shuffled choices.

Form	Q1	Q2	Q3	Difficulty order
QJ2H	D	G	C	easy / medium / hard
QNLW	G	D	A	easy / medium / hard
QKMC	A	E	B	easy / medium / hard
Q2UP	E	B	F	easy / medium / hard
QYU7	C	F	E	easy / medium / hard

Q1: Calculate an observed-minus-fitted residual with the correct sign.

Sources: slides/practice-03.qmd: Questions 1, 4 and 13

Q2: Recover SSR from SER and residual degrees of freedom, then ESS from R squared.

Sources: slides/practice-03.qmd: Questions 9 and 10

Q3: Use an unconditional error restriction and group means to separate a causal slope from an observed comparison.

Sources: slides/practice-03.qmd: Questions 19 and 21; slides/lecture-02.qmd: The observed comparison contains two parts

Form QJ2H - worked solutions

Question 1 (easy) - D: 3

The fitted OLS line is predicted $Y = 100 - 2X$. One observation has $X=10$ and $Y=83$. What is its OLS residual?

The fitted value is $100 - 2 \times 10 = 80$. Residual = observed - fitted = $83 - 80 = 3$.

Question 2 (medium) - G: (100; 300)

An OLS regression with an intercept and one regressor uses $n=27$ observations. $SER=2$ and $R^2=0.75$. SSR denotes the sum of squared residuals; ESS denotes explained sum of squares. Which pair gives (SSR ; ESS)?

$SER^2 = SSR/(n-2)$, so $SSR = 25 \times 2^2 = 100$. Since $1 - R^2 = SSR/TSS = 0.25$, $TSS = 4 \times 100 = 400$. $ESS = TSS - SSR = 300$. The decomposition requires the stated OLS intercept.

Question 3 (hard) - C: (3; -1)

In the constant-effect causal model $Y = \alpha + \beta X + u$, $X=10$ with probability $1/4$ and $X=14$ with probability $3/4$. $E(u)=0$, $E(u | X=10)=3$, and $E(Y | X=14) - E(Y | X=10)=8$. Which pair gives (β ; $E(u | X=14)$)?

Let $m = E(u | X=14)$. Iterated expectations give $0 = (1/4)3 + (3/4)m$, so $m = -1$. The observed mean difference satisfies $8 = \beta(14-10) + (-1-3) = 4\beta - 4$. Hence $\beta = 3$. The observed slope is $8/4 = 2$. The causal interpretation follows from the stated structural model and supplied error restrictions; $E(u)=0$ alone would not identify it from the observed comparison.

Form QNLW - worked solutions

Question 1 (easy) - G: 4

The fitted OLS line is predicted $Y = 110 - 3X$. One observation has $X=10$ and $Y=84$. What is its OLS residual?

The fitted value is $110 - 3 \times 10 = 80$. Residual = observed - fitted = $84 - 80 = 4$.

Question 2 (medium) - D: (225; 675)

An OLS regression with an intercept and one regressor uses $n=27$ observations. $SER=3$ and $R^2=0.75$. SSR denotes the sum of squared residuals; ESS denotes explained sum of squares. Which pair gives (SSR ; ESS)?

$SER^2 = SSR/(n-2)$, so $SSR = 25 \times 3^2 = 225$. Since $1 - R^2 = SSR/TSS = 0.25$, $TSS = 4 \times 225 = 900$. $ESS = TSS - SSR = 675$. The decomposition requires the stated OLS intercept.

Question 3 (hard) - A: (4; -2)

In the constant-effect causal model $Y = \alpha + \beta X + u$, $X=10$ with probability $1/4$ and $X=14$ with probability $3/4$. $E(u)=0$, $E(u | X=10)=6$, and $E(Y | X=14) - E(Y | X=10)=8$. Which pair gives (β ; $E(u | X=14)$)?

Let $m = E(u | X=14)$. Iterated expectations give $0 = (1/4)6 + (3/4)m$, so $m = -2$. The observed mean difference satisfies $8 = \beta(14-10) + (-2-6) = 4\beta - 8$. Hence $\beta = 4$. The observed slope is $8/4 = 2$. The causal interpretation follows from the stated structural model and supplied error restrictions; $E(u)=0$ alone would not identify it from the observed comparison.

Form QKMC - worked solutions

Question 1 (easy) - A: 5

The fitted OLS line is predicted $Y = 120 - 4X$. One observation has $X=10$ and $Y=85$. What is its OLS residual?

The fitted value is $120 - 4 \times 10 = 80$. Residual = observed - fitted = $85 - 80 = 5$.

Question 2 (medium) - E: (400; 1200)

An OLS regression with an intercept and one regressor uses $n=27$ observations. $SER=4$ and $R^2=0.75$. SSR denotes the sum of squared residuals; ESS denotes explained sum of squares. Which pair gives (SSR ; ESS)?

$SER^2 = SSR/(n-2)$, so $SSR = 25 \times 4^2 = 400$. Since $1 - R^2 = SSR/TSS = 0.25$, $TSS = 4 \times 400 = 1600$. $ESS = TSS - SSR = 1200$. The decomposition requires the stated OLS intercept.

Question 3 (hard) - B: (5; -3)

In the constant-effect causal model $Y = \alpha + \beta X + u$, $X=10$ with probability $1/4$ and $X=14$ with probability $3/4$. $E(u)=0$, $E(u | X=10)=9$, and $E(Y | X=14) - E(Y | X=10)=8$. Which pair gives (β ; $E(u | X=14)$)?

Let $m = E(u | X=14)$. Iterated expectations give $0 = (1/4)9 + (3/4)m$, so $m = -3$. The observed mean difference satisfies $8 = \beta(14-10) + (-3-9) = 4\beta - 12$. Hence $\beta = 5$. The observed slope is $8/4 = 2$. The causal interpretation follows from the stated structural model and supplied error restrictions; $E(u)=0$ alone would not identify it from the observed comparison.

Form Q2UP - worked solutions

Question 1 (easy) - E: 6

The fitted OLS line is predicted $Y = 130 - 5X$. One observation has $X=10$ and $Y=86$. What is its OLS residual?

The fitted value is $130 - 5 \times 10 = 80$. Residual = observed - fitted = $86 - 80 = 6$.

Question 2 (medium) - B: (625; 1875)

An OLS regression with an intercept and one regressor uses $n=27$ observations. $SER=5$ and $R^2=0.75$. SSR denotes the sum of squared residuals; ESS denotes explained sum of squares. Which pair gives (SSR ; ESS)?

$SER^2 = SSR/(n-2)$, so $SSR = 25 \times 5^2 = 625$. Since $1 - R^2 = SSR/TSS = 0.25$, $TSS = 4 \times 625 = 2500$. $ESS = TSS - SSR = 1875$. The decomposition requires the stated OLS intercept.

Question 3 (hard) - F: (6; -4)

In the constant-effect causal model $Y = \alpha + \beta X + u$, $X=10$ with probability $1/4$ and $X=14$ with probability $3/4$. $E(u)=0$, $E(u | X=10)=12$, and $E(Y | X=14) - E(Y | X=10)=8$. Which pair gives (β ; $E(u | X=14)$)?

Let $m = E(u | X=14)$. Iterated expectations give $0 = (1/4)12 + (3/4)m$, so $m = -4$. The observed mean difference satisfies $8 = \beta(14-10) + (-4-12) = 4\beta - 16$. Hence $\beta = 6$. The observed slope is $8/4 = 2$. The causal interpretation follows from the stated structural model and supplied error restrictions; $E(u)=0$ alone would not identify it from the observed comparison.

Form QYU7 - worked solutions

Question 1 (easy) - C: 7

The fitted OLS line is predicted $Y = 140 - 6X$. One observation has $X=10$ and $Y=87$. What is its OLS residual?

The fitted value is $140 - 6 \times 10 = 80$. Residual = observed - fitted = $87 - 80 = 7$.

Question 2 (medium) - F: (900; 2700)

An OLS regression with an intercept and one regressor uses $n=27$ observations. $SER=6$ and $R^2=0.75$. SSR denotes the sum of squared residuals; ESS denotes explained sum of squares. Which pair gives (SSR ; ESS)?

$SER^2 = SSR/(n-2)$, so $SSR = 25 \times 6^2 = 900$. Since $1 - R^2 = SSR/TSS = 0.25$, $TSS = 4 \times 900 = 3600$. $ESS = TSS - SSR = 2700$. The decomposition requires the stated OLS intercept.

Question 3 (hard) - E: (7; -5)

In the constant-effect causal model $Y = \alpha + \beta X + u$, $X=10$ with probability $1/4$ and $X=14$ with probability $3/4$. $E(u)=0$, $E(u | X=10)=15$, and $E(Y | X=14) - E(Y | X=10)=8$. Which pair gives (β ; $E(u | X=14)$)?

Let $m = E(u | X=14)$. Iterated expectations give $0 = (1/4)15 + (3/4)m$, so $m = -5$. The observed mean difference satisfies $8 = \beta(14-10) + (-5-15) = 4\beta - 20$. Hence $\beta = 7$. The observed slope is $8/4 = 2$. The causal interpretation follows from the stated structural model and supplied error restrictions; $E(u)=0$ alone would not identify it from the observed comparison.